

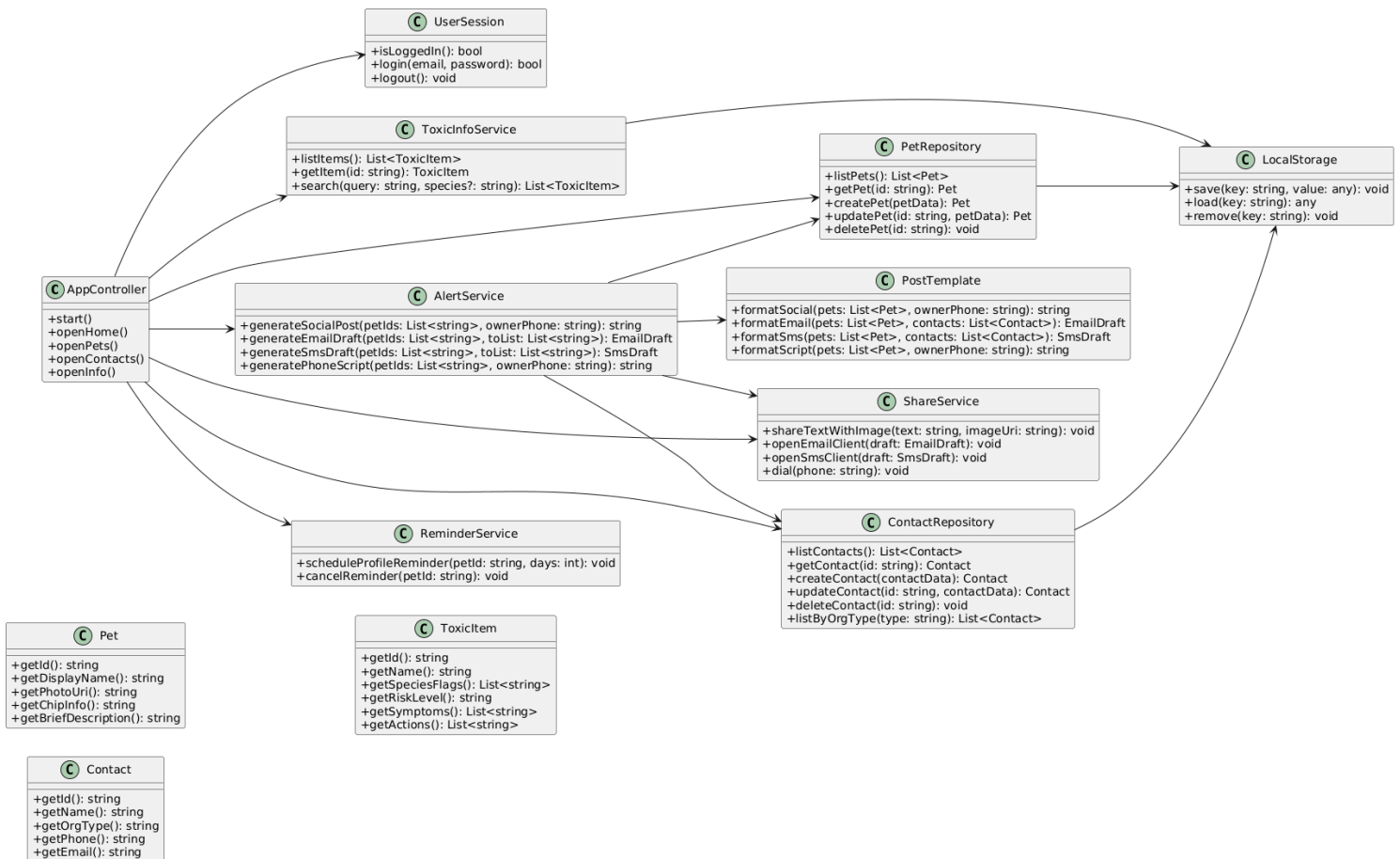
Product Design

Team <Sunrise Inc>

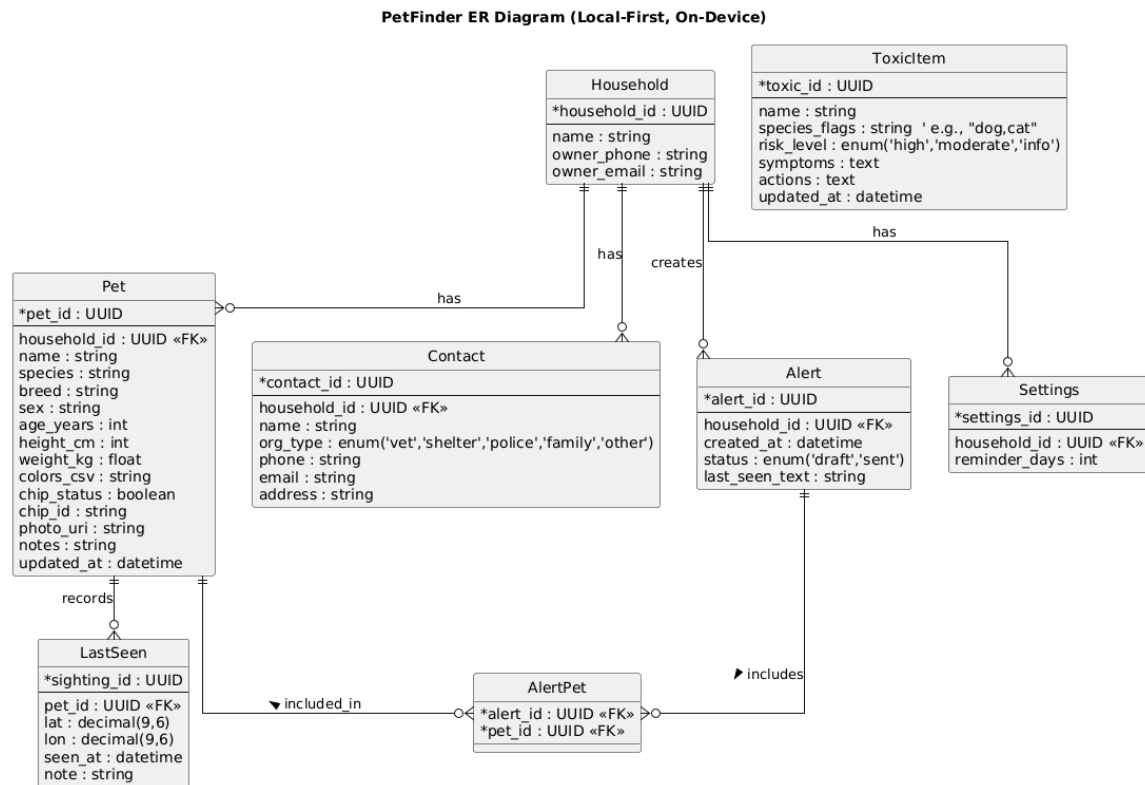
Anjali Fernando, Tanvi Biswal,
Kimberley Juarez, Ojaswi
Subedi, Bibek Pandey

<i>Revision Number</i>	<i>Revision Date</i>	<i>Summary of Changes</i>	<i>Author(s)</i>
0.1	10/08/2025	Initial design document information added	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey
0.2	11/08/2025	Screenshot added from the actual implementation in android emulator. New design summary and design rationale added.	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey
0.3	12/02/2025	Updated UI screenshots, added Login/Signup designs, updated Information Architecture to include Firebase Authentication, and added the Missing Alerts module	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey
0.4	02/14/2026	Migrated pet data to Firebase Firestore & Storage, added microchip field, updated Information Architecture for cloud synchronization	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey
0.5	03/14/2026	Added Reminders module, completed Tips & Toxic Foods feature, improved integration between Contacts and Missing Alerts, and updated UI screenshots to reflect new features	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey

Class Diagram(s)



ER Diagram(s)



Information Architecture Diagram

PetAlert originally followed a fully local-first architecture in Sprint 2. In Sprint 3, the architecture was expanded to include Firebase Authentication. In Sprint 4, the system was migrated to a cloud-based architecture using Firebase Firestore and Firebase Storage. Pet profiles, images, and microchip information are securely stored in Firestore and synchronized across devices.

In Sprint 5, the architecture was further expanded by integrating additional modules including the Reminders system and the completed Tips & Toxic Foods feature. Contacts are now more tightly integrated with the Missing Alerts module, allowing users to select stored contacts when creating alerts. These improvements enhance module interaction while maintaining the modular structure of the system.

Screenshot(s)

Added signup

Android Emulator - Medium_Phone_API_35:5554

7:18

← Create account



Join PetAlert

Email

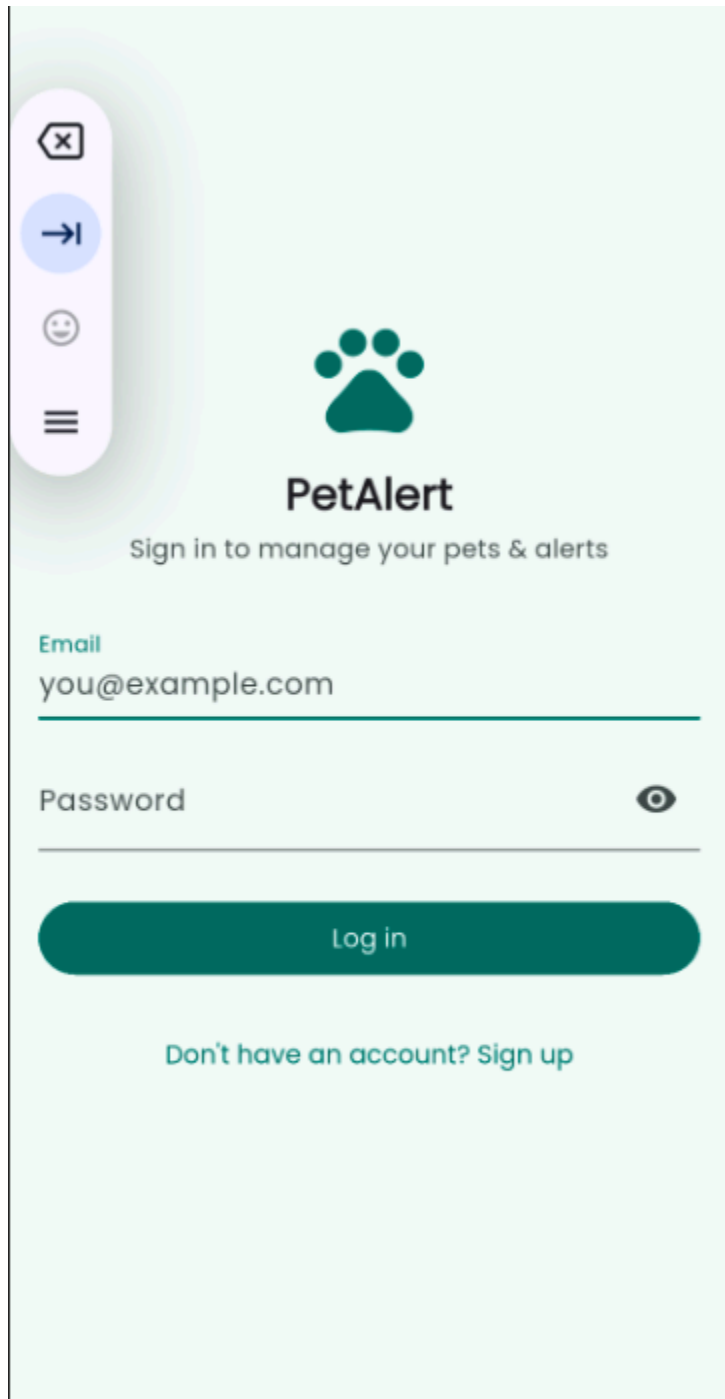
Password 

At least 6 characters

Confirm password 

Create account

Added Login



A mobile app login screen for 'PetAlert'. The background is a light mint green. On the left, there is a vertical white pill-shaped menu with four icons: a house with an 'x', a blue circle with a right arrow, a smiley face, and a hamburger menu. In the center, there is a dark green paw print icon above the text 'PetAlert' in a bold, dark green font. Below the name is the text 'Sign in to manage your pets & alerts' in a smaller, grey font. The login form consists of two fields: 'Email' with the placeholder text 'you@example.com' and 'Password' with a toggle eye icon on the right. Below the fields is a large, rounded dark green button with the text 'Log in' in white. At the bottom, there is a link that says 'Don't have an account? Sign up' in a teal color.







PetAlert
Sign in to manage your pets & alerts

Email
you@example.com

Password 

Log in

Don't have an account? Sign up

Creating Missing Alert

 Create Missing Alert

Select pet

pupp

Last seen location

Last seen date & time

Not set



Contact name (optional)

Contact phone (optional)

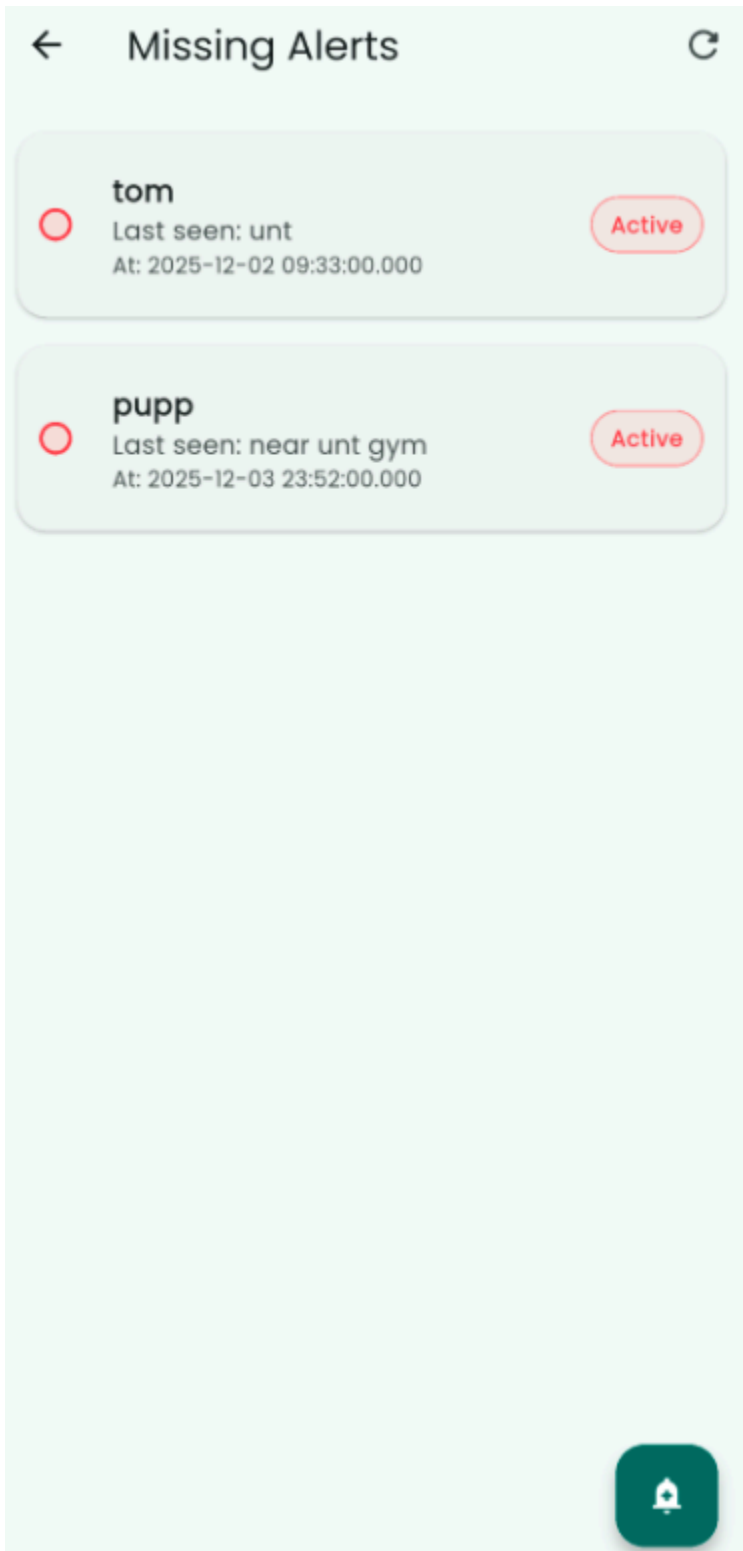
Notes (optional)

Status

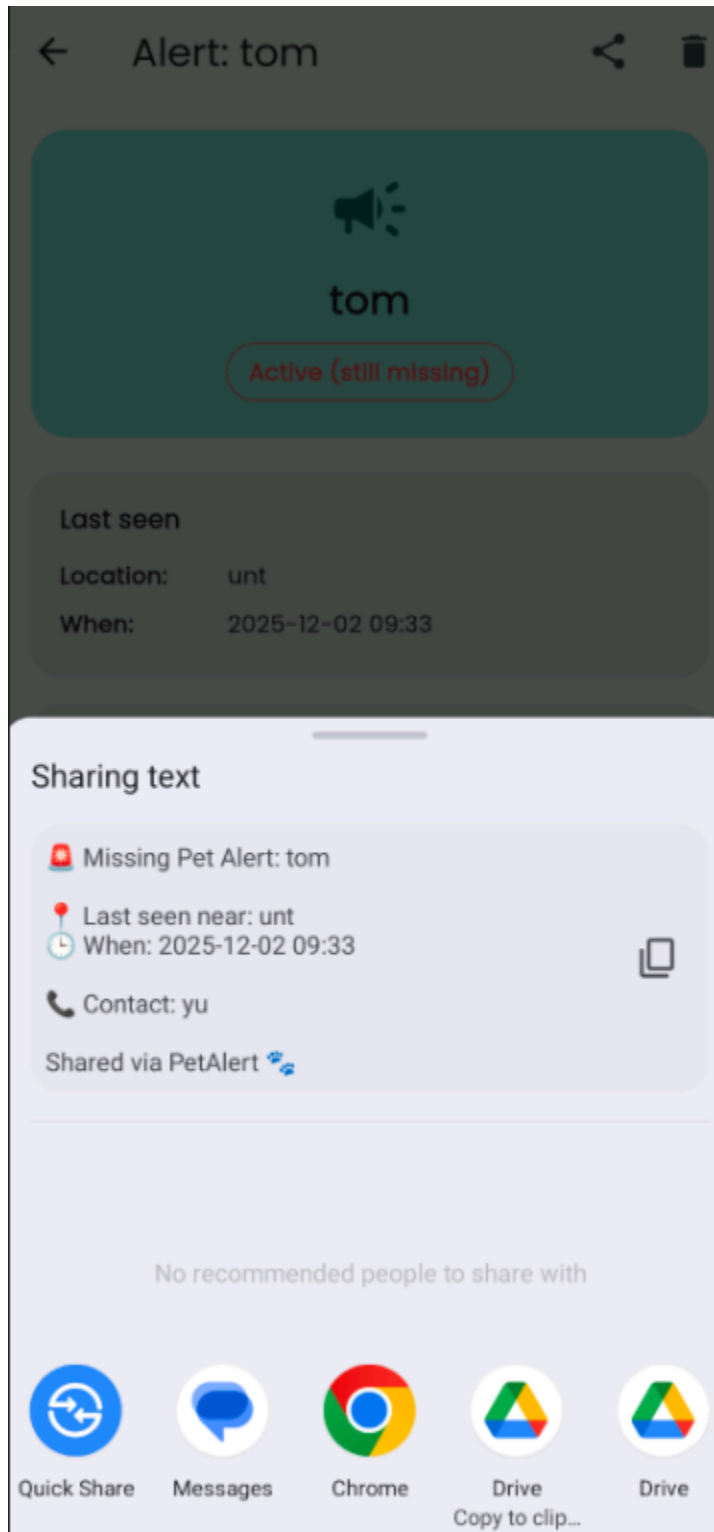
Active (still missing)

 Save Alert

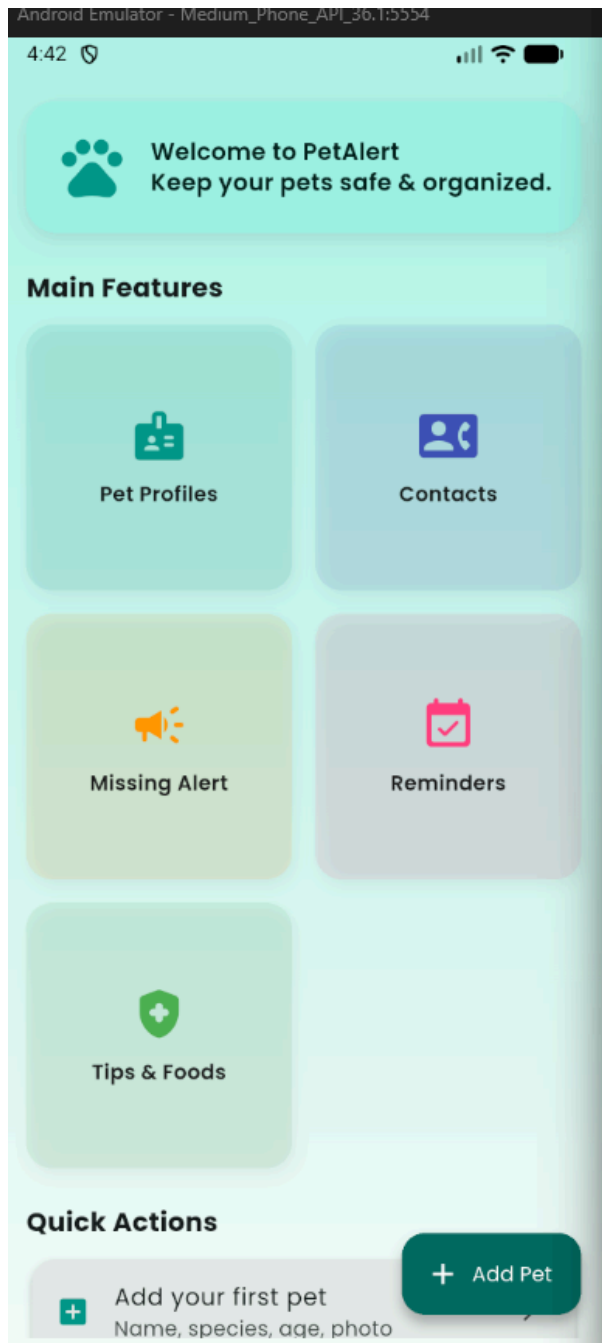
Missing Alert List



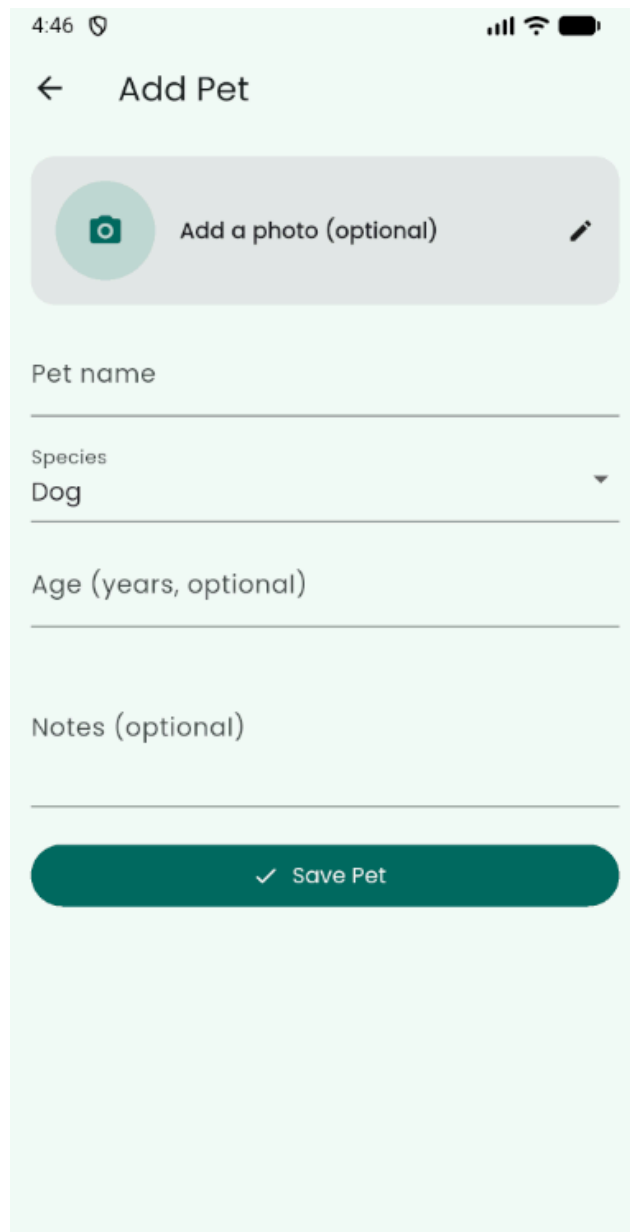
Sharing Missing Alert



Homescreen is loaded in the android emulator.



Add pet feature:



A mobile application interface for adding a new pet. The screen has a light green background. At the top, the status bar shows the time 4:46, signal strength, Wi-Fi, and battery. Below the status bar is a header with a back arrow and the text 'Add Pet'. The main form area contains several input fields: a photo upload section with a camera icon and the text 'Add a photo (optional)'; a text field for 'Pet name'; a dropdown menu for 'Species' with 'Dog' selected; a text field for 'Age (years, optional)'; and a text field for 'Notes (optional)'. At the bottom is a large green button with a white checkmark and the text 'Save Pet'.

4:46

← Add Pet

Add a photo (optional)

Pet name

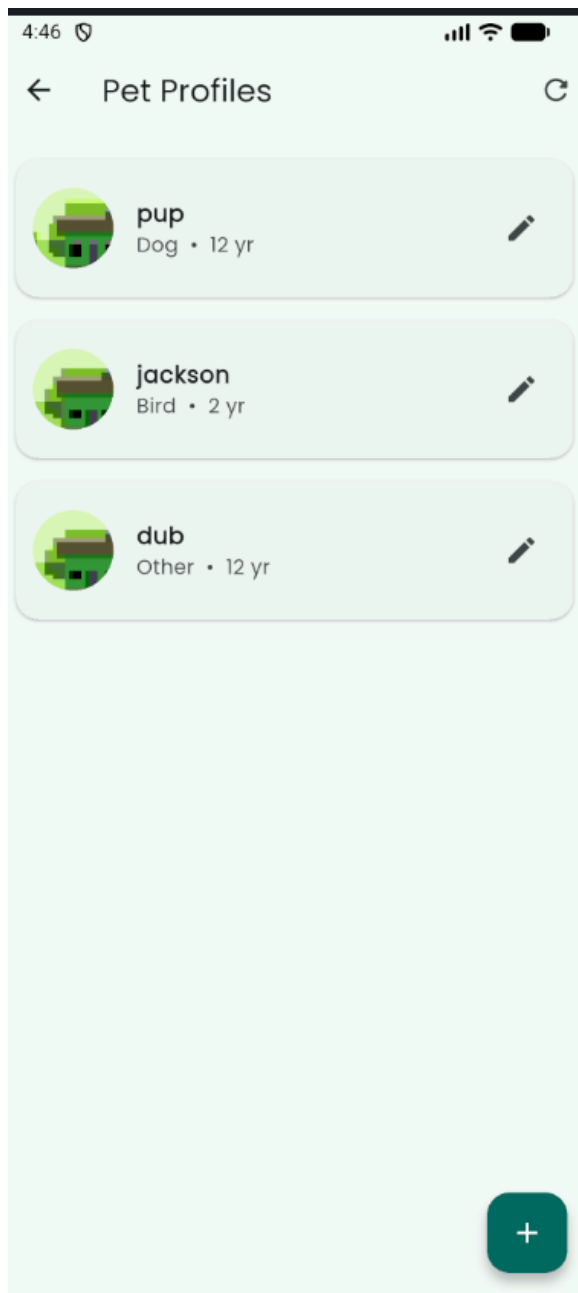
Species
Dog

Age (years, optional)

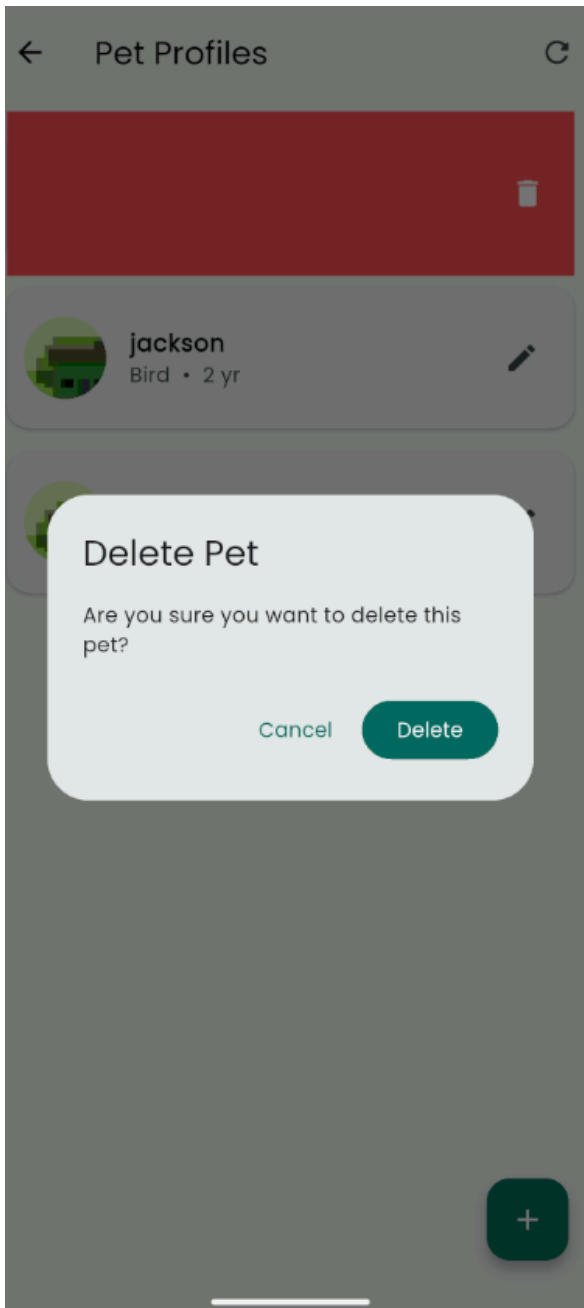
Notes (optional)

✓ Save Pet

Pet Profiles feature:



Delete profile feature



We added the microchip option in missing alert screen

Android Emulator - Medium_Phone_API_36.1:5554

11:03

← Create Missing Alert

Last seen location

Last seen date & time
Not set

Calendar icon

Contact name (optional)

Contact phone (optional)

Notes (optional)

Optional details

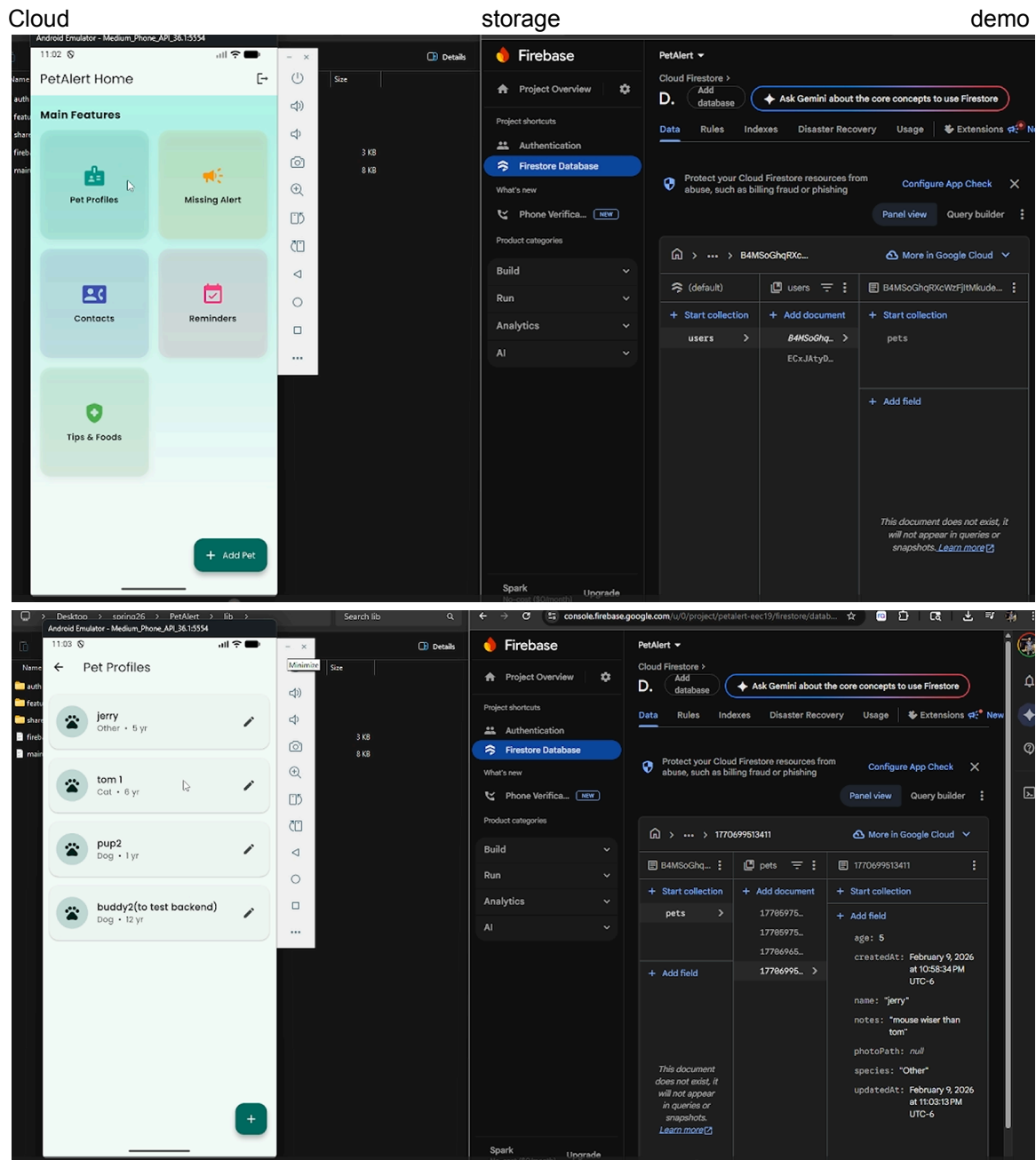
Microchip ID, distinguishing marks

Microchip ID (optional)
e.g., 9851210XXXXXX

Distinguishing marks (optional)

Status
Active (still missing)

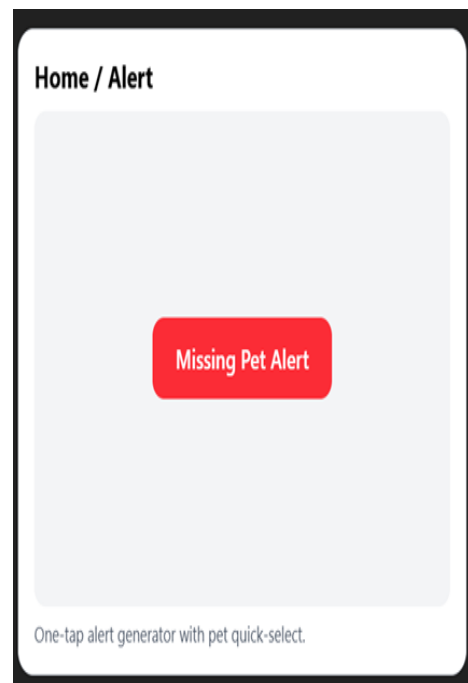
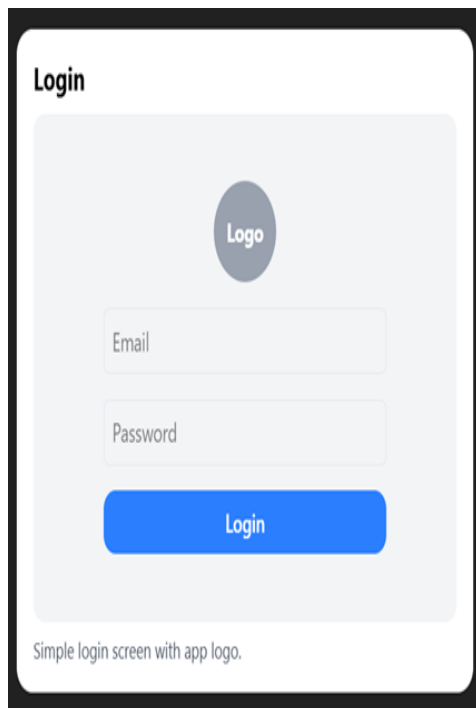
✓ Save Alert



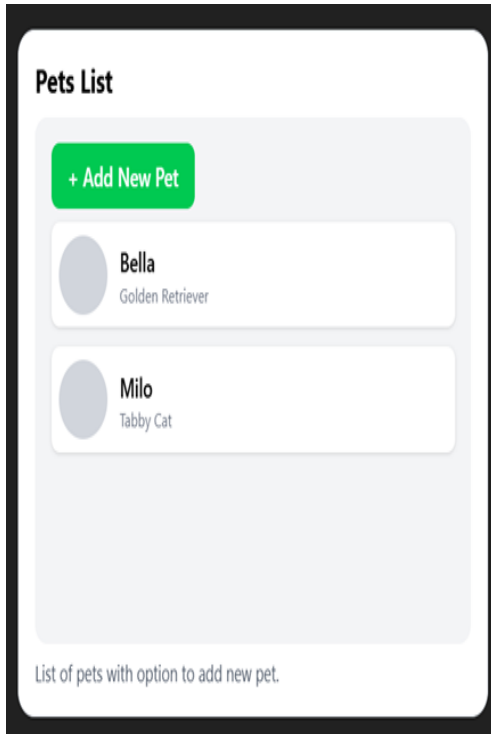
User Interface Wireframe(s)

This is the login page where the user logs in to the PetAlert app

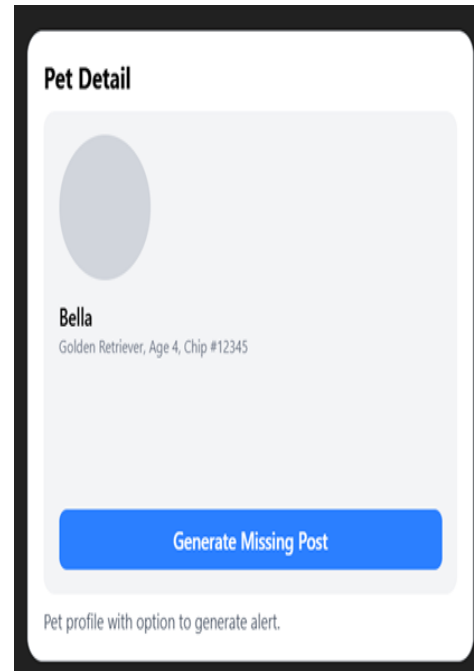
This is a missing pet alert page, where a user needs to report a missing pet by clicking the button



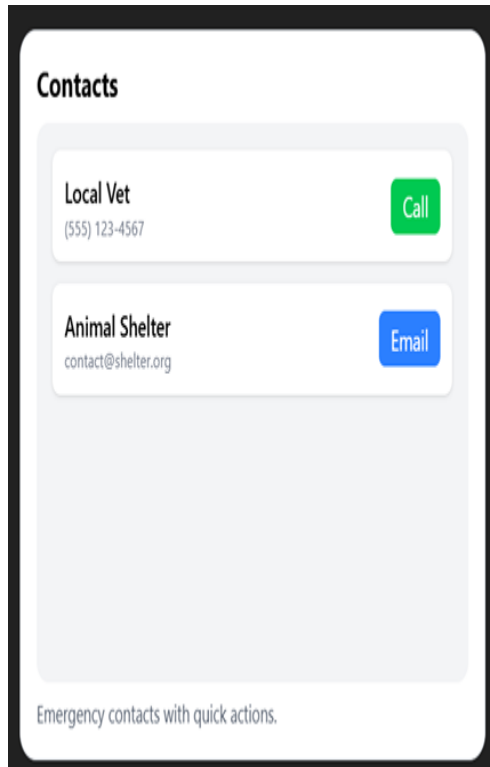
This displays a list of pets, that the user inputs. If the user wants to add a new pet there is a green button above.



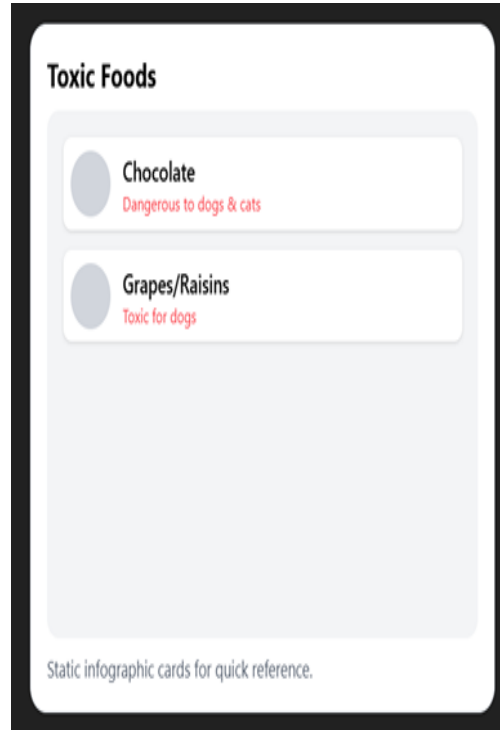
This is a pet profile page where it shows the pet account info



This page lists all the emergency contacts for that specific pet in case they need help



This page will show the infographic of toxic foods that pets can't eat



Design Summary

The design of PetAlert continued to evolve in Sprint 5 with the addition of several new functional modules and improved integration between existing components. While Sprint 4 focused on migrating the system from local storage to a cloud-based architecture using Firebase Firestore and Firebase Storage, Sprint 5 expanded the application's capabilities to provide more comprehensive pet management tools.

The reminders module was introduced to allow users to schedule and manage important pet-related events such as veterinary visits, vaccinations, or medication reminders. This feature currently uses local storage to ensure quick access and reliability.

The Tips & Toxic Foods feature was also completed in this sprint, providing users with categorized information about foods that are dangerous or safe for pets. This improves the educational value of the application and helps pet owners make safer decisions.

Additionally, integration between the Contacts module and the Missing Alerts system was improved. Users can now select saved contacts when generating a missing pet alert, reducing redundant data entry and improving workflow efficiency.

These enhancements extend the functionality of PetAlert while preserving the clean interface and emergency-focused design philosophy established in earlier sprints.

Design Rationale

The design decisions in Sprint 5 focused on improving usability, expanding system functionality, and strengthening interaction between application modules. Adding the reminders feature improves the practical value of the application by helping users keep track of important pet care tasks directly within the app.

Completing the Tips & Toxic Foods module aligns with the goal of supporting pet owners not only during emergency situations but also in everyday pet care. Providing easily accessible safety information helps users quickly identify harmful foods and respond appropriately.

Improving the integration between contacts and missing alerts simplifies the process of generating alerts and reduces repetitive user input. This strengthens the internal consistency of the system and improves overall user experience.

These updates maintain the modular architecture of the application while continuing to prioritize fast access to critical pet information and an intuitive interface.